

FSD Documentation Format

Full Stack Development (FSD) Project Documentation

shopEZ E-commerce Marketplace | MERN Stack | June 2026

1. Project Overview

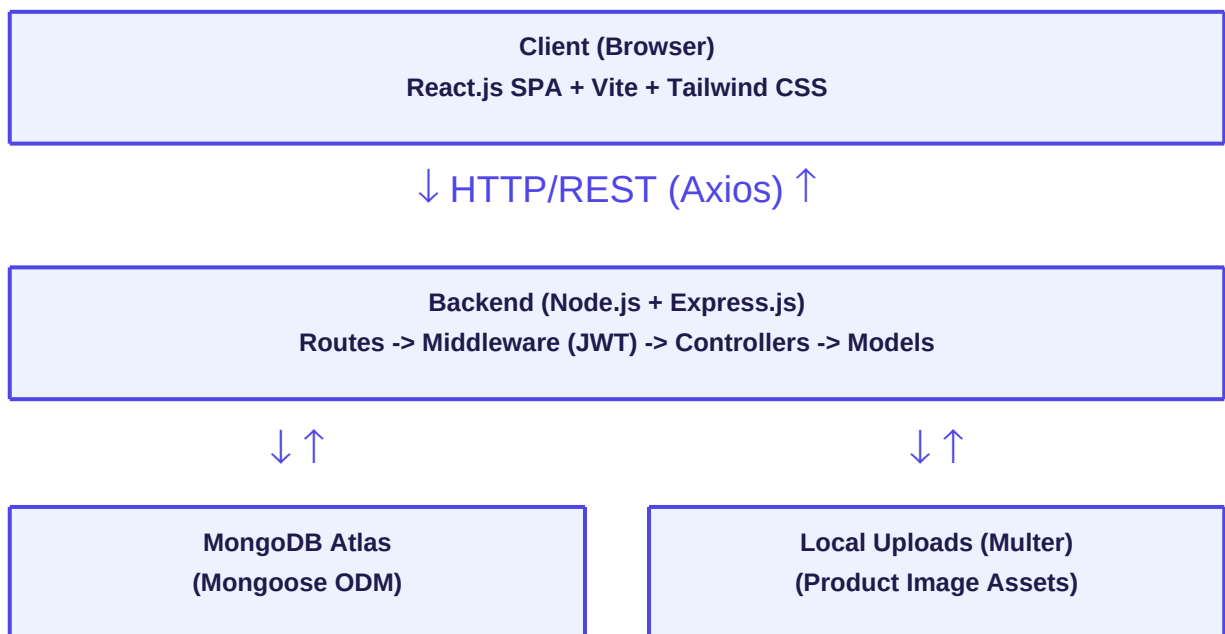
shopEZ is a fully functional, secure, and scalable full-stack e-commerce marketplace platform that delivers a seamless online shopping experience for customers and a powerful administrative control center for store managers and sellers. Built on the MERN stack (MongoDB, Express.js, React.js, Node.js).

Feature Area	Highlights
Authentication	JWT-based sessions, Bcrypt password hashing, Role-Based Access Control (Customer / Seller / Admin)
Product Catalog	Live autocomplete search, multi-parameter filters (price, category, rating, stock), sort options
Shopping Cart	Persistent cross-session cart, stock validation, real-time totals

Wishlist	Save favorites, one-click Move to Cart with auto-clear
Checkout	Multi-step wizard (Address, Shipping, Coupon, Payment), itemized billing
Payments	Mock Stripe Payment Intent endpoint simulating production-grade integration
Reviews	Verified purchase badge, star ratings, helpful upvote tallies
Order Tracking	6-step fulfillment timeline (Placed, Confirmed, Packed, Shipped, Out for Delivery, Delivered)
Admin Dashboard	Chart.js analytics, product/user/order/coupon CRUD, low-stock alerts

2. Architecture Overview

shopEZ implements the Model-View-Controller (MVC) pattern across a fully decoupled MERN stack:



3. Technology Stack

Layer	Technology	Version	Justification
Frontend Framework	React.js	19.x	Component-driven SPA with virtual DOM
Build Tooling	Vite	8.x	Lightning-fast HMR with optimized builds
Styling	Tailwind CSS	3.x	Utility-first responsive design
Routing (FE)	React Router DOM	6.x	Declarative route hierarchy with protected guards
HTTP Client	Axios	1.7.x	Promise-based HTTP with interceptors
Charts	Chart.js + react-chartjs-2	4.x	Canvas-rendered analytics charts
Icons	Lucide React	0.395.x	Accessible SVG icon library
Backend Runtime	Node.js	18+	Server-side JavaScript
Web Framework	Express.js	4.x	Minimal RESTful API framework
Database	MongoDB + Mongoose	8.x	NoSQL document store with schema validation

Authentication	JWT + Bcrypt.js	Latest	Stateless token sessions + password hashing
Image Storage	Local Multer	1.4.x	Disk-based image persistence
Dev Fallback DB	mongodb-memory-server	11.x	In-memory MongoDB for development

4. Prerequisites

- **Node.js** v18 or above — nodejs.org
- **npm** v8+ (bundled with Node.js)
- **MongoDB** (Local Instance OR MongoDB Atlas cloud cluster)
- **Git** for version control

5. Installation and Setup Guide

Step 1: Clone the Repository

```
git clone https://github.com/yrkoppu17/E-Commerce-Application.git
cd "E-Commerce Application"
```

Step 2: Configure Backend Environment Variables

Navigate to the backend/ directory and create a .env file from the template:

```
cd backend
copy .env.example .env
```

Open .env and fill in your values:

Variable	Example Value	Description
PORT	5000	Backend server port

MONGO_URI	mongodb://127.0.0.1:27017/shopez	MongoDB connection string
JWT_SECRET	your_super_secret_key	JWT signing secret
STRIPE_API_KEY	your_stripe_key	Stripe API key (mock)

Note: If MONGO_URI is unreachable, the seeder automatically starts a local mongodb-memory-server in-memory fallback and seeds it with sample customers, products, reviews, and active coupons.

Step 3: Install Backend Dependencies

```
cd backend && npm install
```

Step 4: Install Frontend Dependencies

```
cd frontend && npm install
```

Step 5 (Optional): Seed the Database

```
cd backend && npm run seed
```

Seeds: admin account, customer accounts, 20+ products across multiple categories, active promotional coupon codes, and sample reviews.

6. Running the Application

Terminal 1: Backend API Server

```
cd backend && npm run dev — Server starts on http://localhost:5000
```

Terminal 2: Frontend Dev Server

```
cd frontend && npm run dev — App starts on http://localhost:5173
```

7. Folder Structure

Backend (/backend)

Path	Description
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config/db.js	MongoDB connection with in-memory fallback
controllers/authController.js	User register, login, profile, wishlist, addresses
controllers/productController.js	Catalog search, reviews, helpful votes
controllers/orderController.js	Order creation, payment update, delivery milestones
controllers/couponController.js	Coupon CRUD, active validation
middleware/authMiddleware.js	JWT protect(), admin(), seller() guards
middleware/errorMiddleware.js	Centralized notFound and errorHandler
models/User.js	User schema (addresses[], wishlist[], role)
models/Product.js	Product schema (reviews[], stock, rating)
models/Order.js	Order schema (items, payment, delivery timeline)
models/Coupon.js	Coupon schema (code, type, value, expiry)
routes/authRoutes.js	/api/users endpoints
routes/productRoutes.js	/api/products endpoints
routes/orderRoutes.js	/api/orders endpoints
routes/couponRoutes.js	/api/coupons endpoints
routes/uploadRoutes.js	/api/upload (Multer file upload)
utils/seeder.js	Database seed script
uploads/	Local product image storage

server.js	Express app entry point
.env.example	Environment variable template

Frontend (/frontend)

Path	Description
src/assets/	Static assets (logos, icons)
src/components/Navbar.jsx	Top navigation with cart badge, auth links
src/components/ProductCard.jsx	Reusable product listing card
src/components/ProtectedRoute.jsx	Route guard (auth + role verification)
src/context/AuthContext.jsx	User session state and dispatch
src/context/CartContext.jsx	Shopping cart state (items, totals)
src/context/ToastContext.jsx	Global toast notification system
src/pages/Home.jsx	Catalog browse with search and filters
src/pages/ProductDetails.jsx	Single product, reviews, related carousel
src/pages/Cart.jsx	Cart summary with quantity controls
src/pages/Wishlist.jsx	Saved wishlist with move-to-cart
src/pages/Checkout.jsx	Multi-step wizard with coupon and mock payment
src/pages/MyOrders.jsx	Order history with fulfillment tracker
src/pages/ProfileDashboard.jsx	Edit profile, manage addresses

src/pages/AdminDashboard.jsx	Admin panel (analytics, CRUD, coupons)
src/pages/SellerDashboard.jsx	Seller panel (products, orders)
src/pages/Login.jsx	Sign-in form
src/pages/Register.jsx	Registration form
index.html	Vite HTML template
vite.config.js	Vite configuration

8. RESTful API Documentation

All API routes served from `http://localhost:5000`. Protected routes require: `Authorization: Bearer <jwt_token>`

8.1 Authentication Routes: `/api/users`

Method	Endpoint	Auth	Description
POST	<code>/api/users</code>	Public	Register a new user account
POST	<code>/api/users/login</code>	Public	Log in and retrieve JWT token
GET	<code>/api/users/profile</code>	User	Get current user profile
PUT	<code>/api/users/profile</code>	User	Update name, email, or password
GET	<code>/api/users/wishlist</code>	User	Retrieve user wishlist products

POST	/api/users/wishlist/:productId	User	Toggle product in/out of wishlist
POST	/api/users/addresses	User	Add a new shipping address
PUT	/api/users/addresses/:addressId	User	Edit an existing shipping address
DELETE	/api/users/addresses/:addressId	User	Remove a shipping address
GET	/api/users	Admin	Get all registered users
PUT	/api/users/:id/block	Admin	Block or unblock a user account

8.2 Product Routes: /api/products

Method	Endpoint	Auth	Description
GET	/api/products	Public	List all products with optional search/filter/sort
GET	/api/products/suggestions	Public	Get autocomplete name suggestions
GET	/api/products/:id	Public	Get single product detail
GET	/api/products/:id/related	Public	Get related product recommendations
POST	/api/products	Seller/Admin	Create a new product listing

PUT	/api/products/:id	Seller/Admin	Update product information
DELETE	/api/products/:id	Seller/Admin	Delete a product listing
POST	/api/products/:id/reviews	User	Submit a product review with star rating
POST	/api/products/:id/reviews/:reviewId/vote	User	Toggle helpful vote on a review

8.3 Order Routes: /api/orders

Method	Endpoint	Auth	Description
POST	/api/orders	User	Place a new order (deducts stock)
GET	/api/orders/myorders	User	Retrieve current user order history
GET	/api/orders/:id	User	Get single order details
PUT	/api/orders/:id/pay	User	Update order payment status to paid
GET	/api/orders	Admin	List all platform orders
PUT	/api/orders/:id/deliver	Admin	Update order delivery status milestone

8.4 Coupon Routes: /api/coupons

Method	Endpoint	Auth	Description
GET	/api/coupons	Admin	List all coupon codes
POST	/api/coupons	Admin	Create a new coupon rule
PUT	/api/coupons/:id	Admin	Edit coupon details or toggle active status
DELETE	/api/coupons/:id	Admin	Delete a coupon permanently
POST	/api/coupons/validate	User	Validate coupon code and retrieve discount

8.5 Payment Route: /api/payment

Method	Endpoint	Auth	Description
POST	/api/payment/create-intent	User	Create a mock Stripe payment intent returning a client secret

8.6 Upload Route: /api/upload

Method	Endpoint	Auth	Description
POST	/api/upload	Seller/Admin	Upload a product image via Multer (stored in /uploads)

9. Authentication and Security Model

- **Password Hashing:** All user passwords are hashed via bcryptjs with a salt round factor of 10 before being persisted to MongoDB.
- **Session Tokens:** Upon successful login, the server signs a JWT containing user._id and user.role with a configurable secret (JWT_SECRET). Tokens carry a 30-day expiry.
- **Protected Routes (Backend):** The protect middleware in authMiddleware.js verifies the Authorization header on every protected endpoint. The admin() and seller() middleware further restrict access based on the decoded role field.
- **Protected Routes (Frontend):** The ProtectedRoute.jsx component wraps sensitive React pages, redirecting unauthenticated users to the Login page.

10. Known Constraints

Payment Integration: The current Stripe integration uses a simulated mock endpoint. A real Stripe publishable key and configured webhook listener are required for live transaction processing.

Image Storage: Product images are stored locally in backend/uploads/. For production deployments, replacing Multer's disk storage with Cloudinary or AWS S3 is recommended.

Seller Profile: Seller onboarding requires admin approval via the admin panel before a seller account gains product management privileges.

11. Future Enhancements

- **Real Payment Gateway Integration:** Wire actual Stripe webhook handling to confirm payment events asynchronously.
- **Cloud Image CDN:** Integrate Cloudinary SDK to replace local Multer disk storage.
- **AI Product Recommendations:** Implement ML-driven collaborative filtering to personalize related-product suggestions.
- **Multi-Vendor Marketplace:** Expand seller onboarding to support independent vendor profiles and commission tracking.
- **Advanced Analytics:** Add time-series graphs, cohort retention charts, and inventory forecasting.
- **Push Notifications:** Notify customers via email or browser push on order status milestones (Shipped, Delivered).